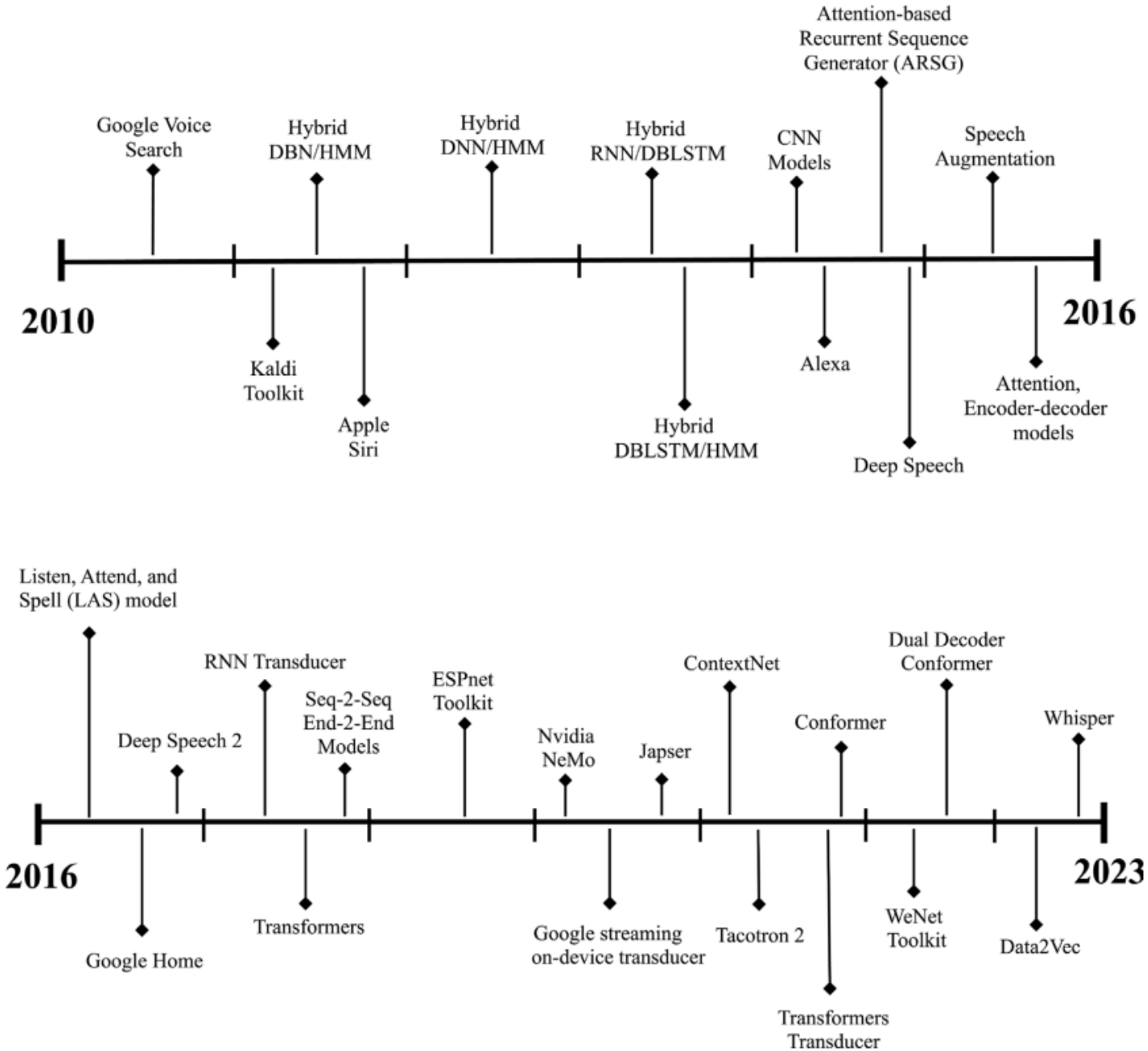


Fig. 4. A timeline of some of the significant advances in speech recognition from 2010 to 2023.

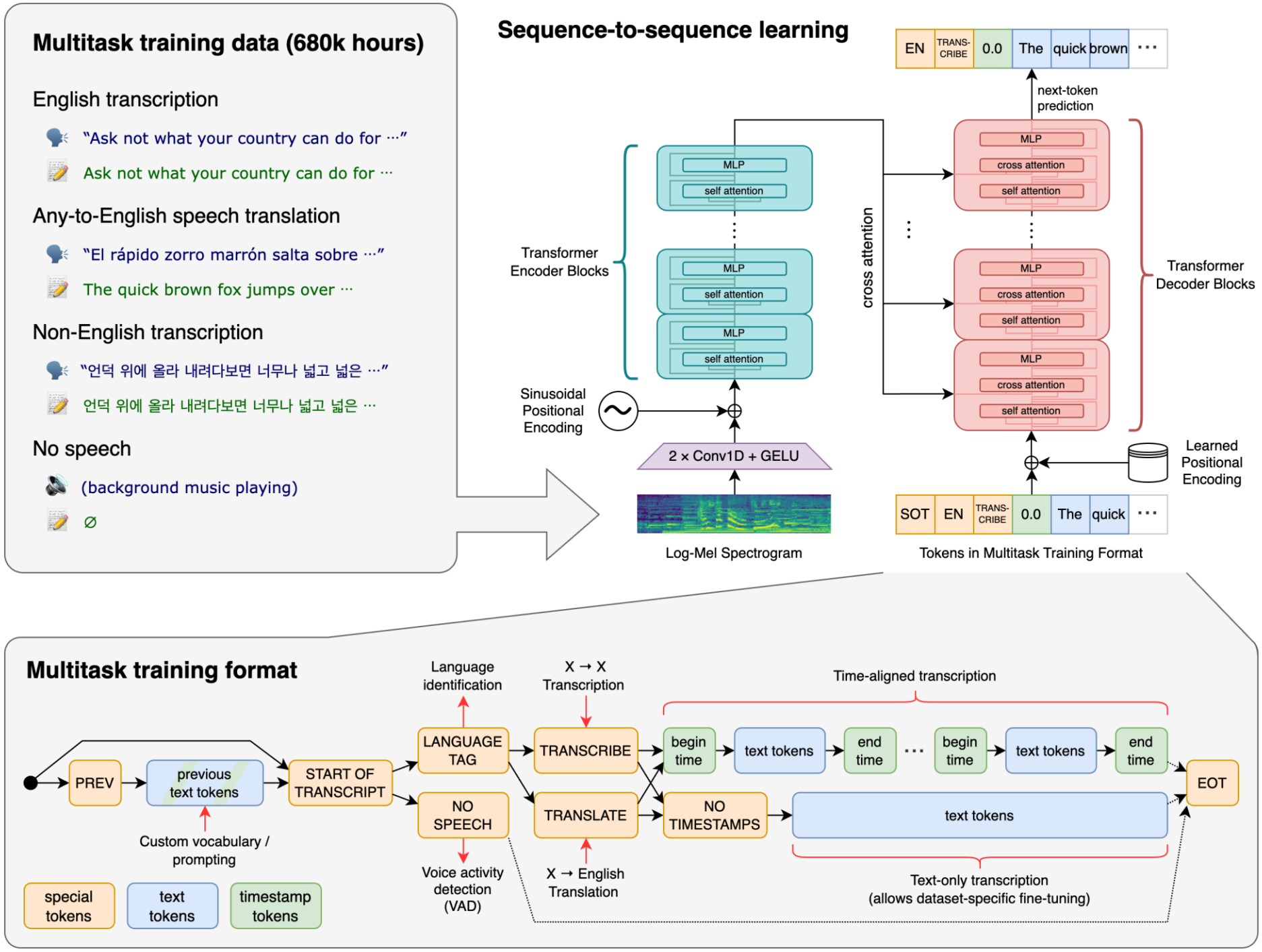
1. HMM-GMM(Hidden Markov Model – Gaussian Mixture Model. 1980s-2000s)
2. DNN (DNN-HMM 2010-2014)
3. CTC 2006
4. RNN-T 2012 streaming neural ASR
5. LAS(Listen, Attend, and Spell): 2016
6. Transformer 2017
7. Conformer 2020 (Attention: models long-range relationships across speech frames.
   Convolution: captures local patterns such as short-term spectral and phonetic structure)

ASR

1. Whisper 2022
2. SenseVoice Tongyi 2024.7
3. Qwen2-Audio 7B 2024.7
4. VibeVoice 7B microsoft 2025
5. VibeVoiceASR 2026.3
6. Kimi-Audio 2025.4
7. Step-Audio 2025.8
8. Mimo-Audio 7B. 2025.12
9. Moss-Transcribe-Diarize 0.9B 2026.7
10. Cohere-transcribe-03-2026
11. Qwen3-ASR-1.7B 2026.1

	Time	Organization	Paper	
Whisper Whisper-turbo 1550M	2022.Sep 2022.Dec	OpenAI	Robust Speech Recognition via Large-Scale Weak Supervision	https://arxiv.org/pdf/2212.04356
SenseVoice 7B	2024.7	Alibaba Group Tongyi SpeechTeam	FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs	https://arxiv.org/pdf/2407.04051
Qwen2-Audio 7B	2024.7	Alibaba	Qwen2-Audio Technical Report	https://arxiv.org/pdf/2407.10759
VibeVoice 7B VibeVoiceASR 7B	2025 2026 3	Microsoft	VibeVoice Technical Report VIBEVOICE-ASR Technical Report	https://arxiv.org/pdf/2508.19205 https://arxiv.org/pdf/2601.18184
Step-Audio	2025.8	StepFun Audio Team	Step-Audio 2 Technical Repor	https://arxiv.org/pdf/2507.16632
Mimo-Audio 7B	2025.12	XiaoMi	MiMo-Audio: Audio Language Models are Few-Shot Learners	https://arxiv.org/pdf/2512.23808
Moss-Transcribe-diarize 0.9B	2026. 7	MOSI.AI	MOSS Transcribe Diarize Technical Report	https://arxiv.org/pdf/2601.01554
Qwen3-ASR 1.7B	2026.1	Qwen	Qwen3-ASR Technical Report	https://arxiv.org/pdf/2601.21337

Robust Speech Recognition via Large-Scale Weak Supervision - 2022



FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs Alibaba Group -07-2024

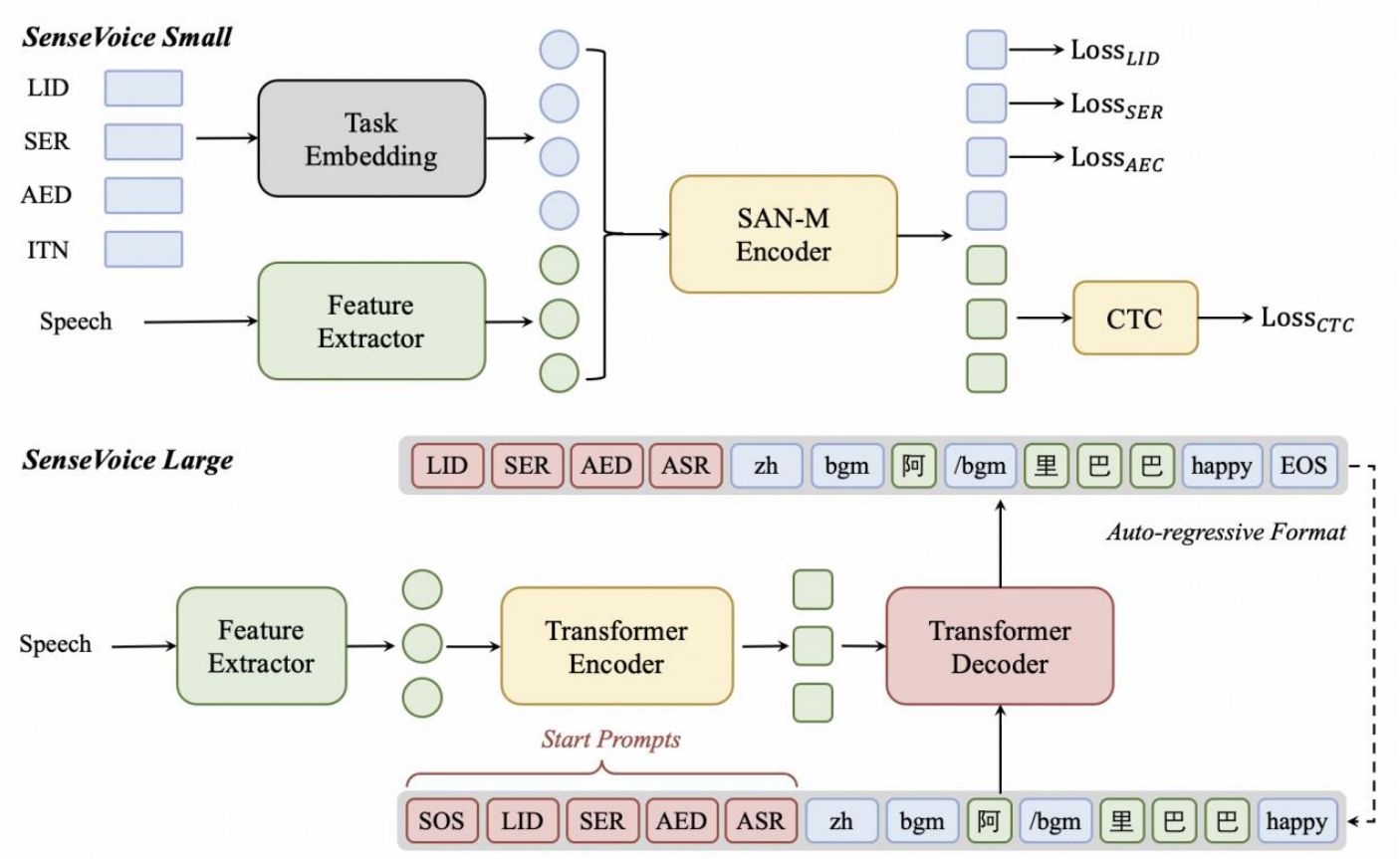


Figure 2: SenseVoice is a comprehensive speech foundation model designed to perform various speech understanding tasks, including Automatic Speech Recognition (ASR), Language Identification (LID), Speech Emotion Recognition (SER), and Audio Event Detection (AED). SenseVoice-Small [Top]: An encoder-only model optimized for rapid speech understanding. It offers high-speed processing while supporting 5 languages. SenseVoice-Large [Bottom]: An encoder-decoder model aimed at achieving more precise speech understanding across a broader range of languages. It excels in accuracy and supports an extensive set of language capabilities.

Qwen2-Audio Technical Report 2024-02-02

Qwen2-Audio-7B

1. Generalization LALM

SLU(Spoken language understanding)
SER(Speech Emotion Recognition)
ASR(Audio Speech Recognition)
S2TT(Speech-to-Text Translation)
VSC(Vocal Sound Classification)

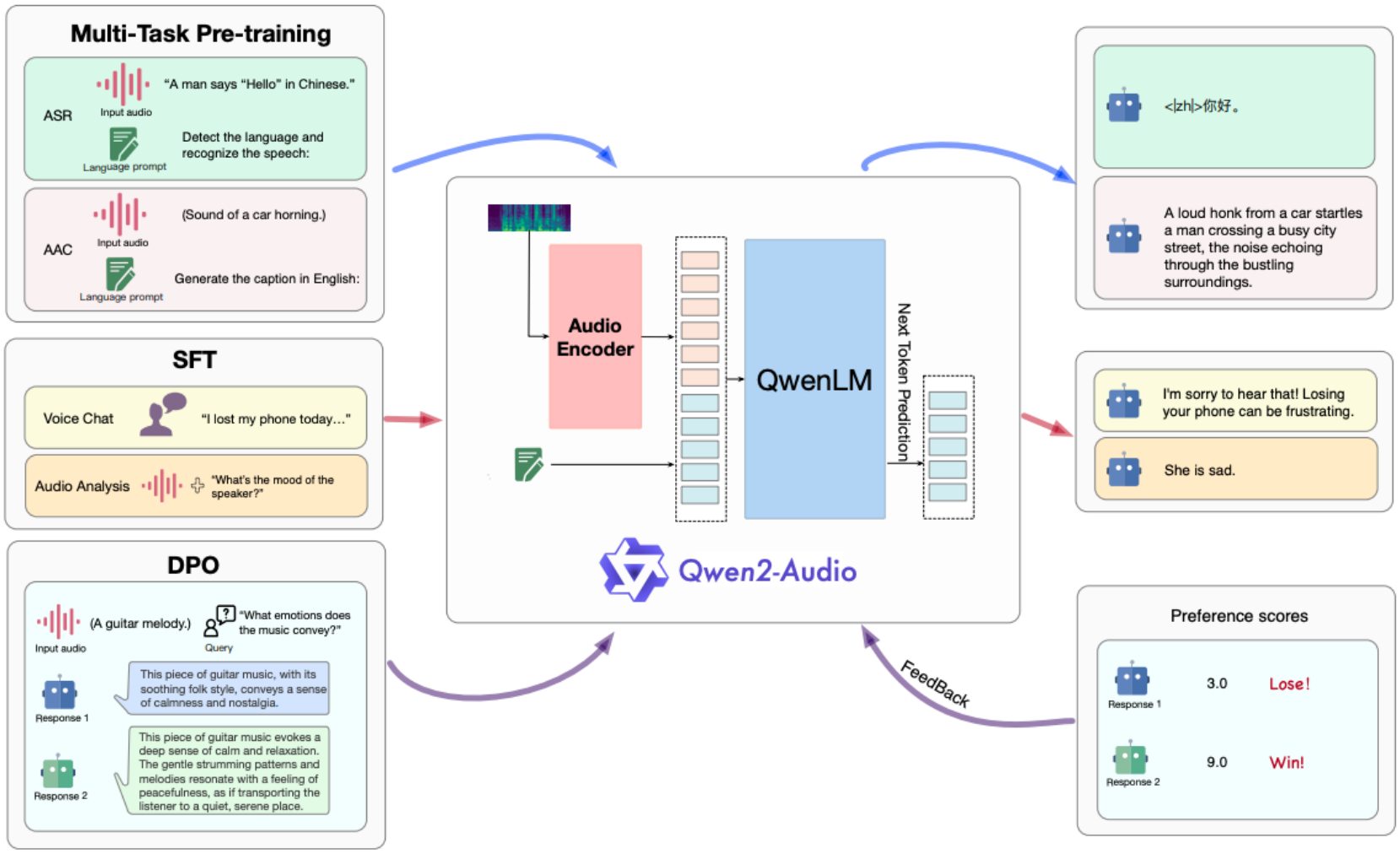


Figure 2: The overview of three-stage training process of Qwen2-Audio.

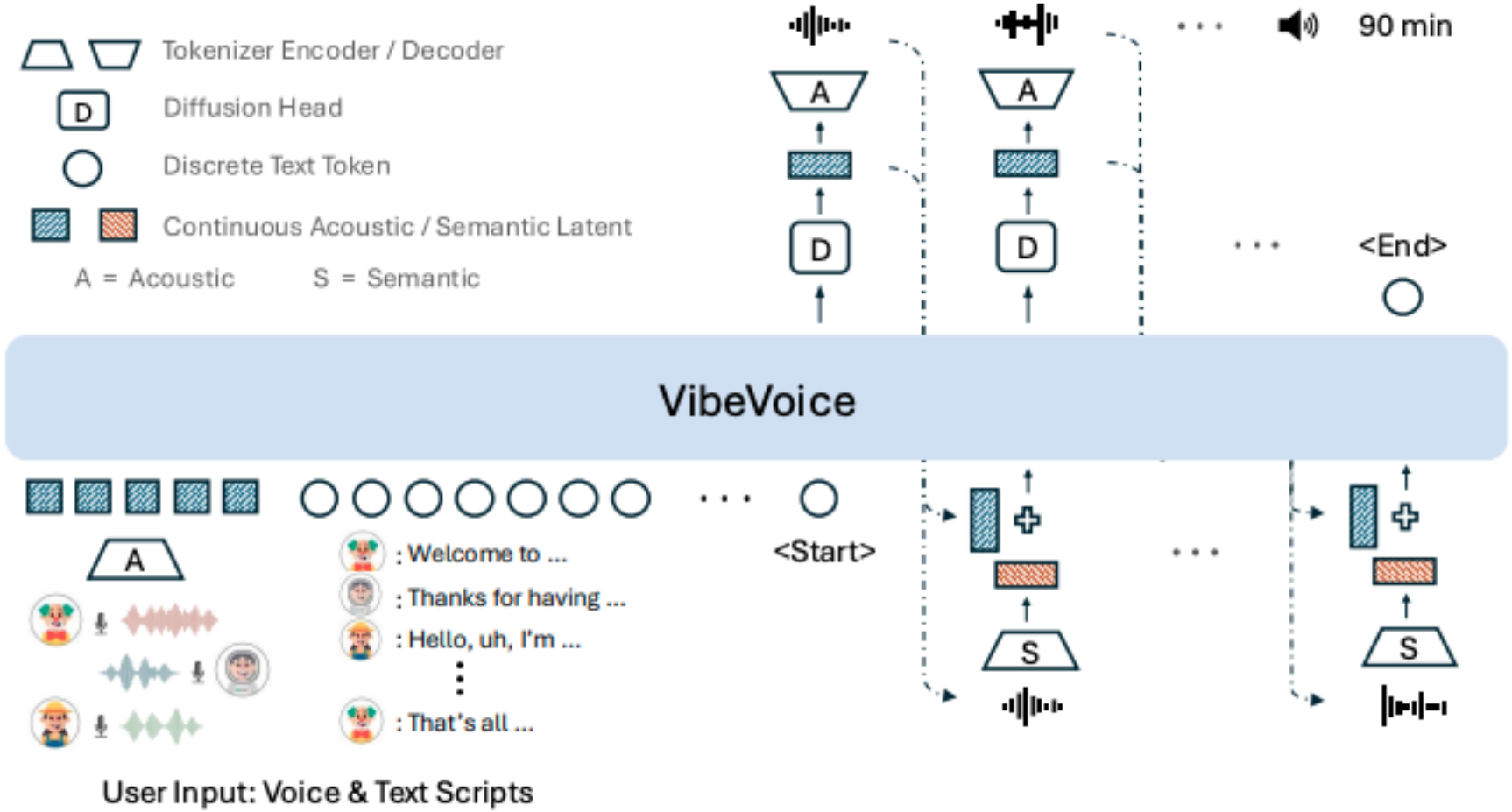


Figure 2: VIBEVOICE employs next token diffusion framework as in LatentLM [SBW+24] to synthesize long-form and multi-speaker audios. Voice prompts and text scripts provide initial input. VIBEVOICE processes hybrid context features, and its hidden states condition a token level Diffusion Head (D), which predicts acoustic VAE for speech segments, subsequently recovered by acoustic decoder (A).

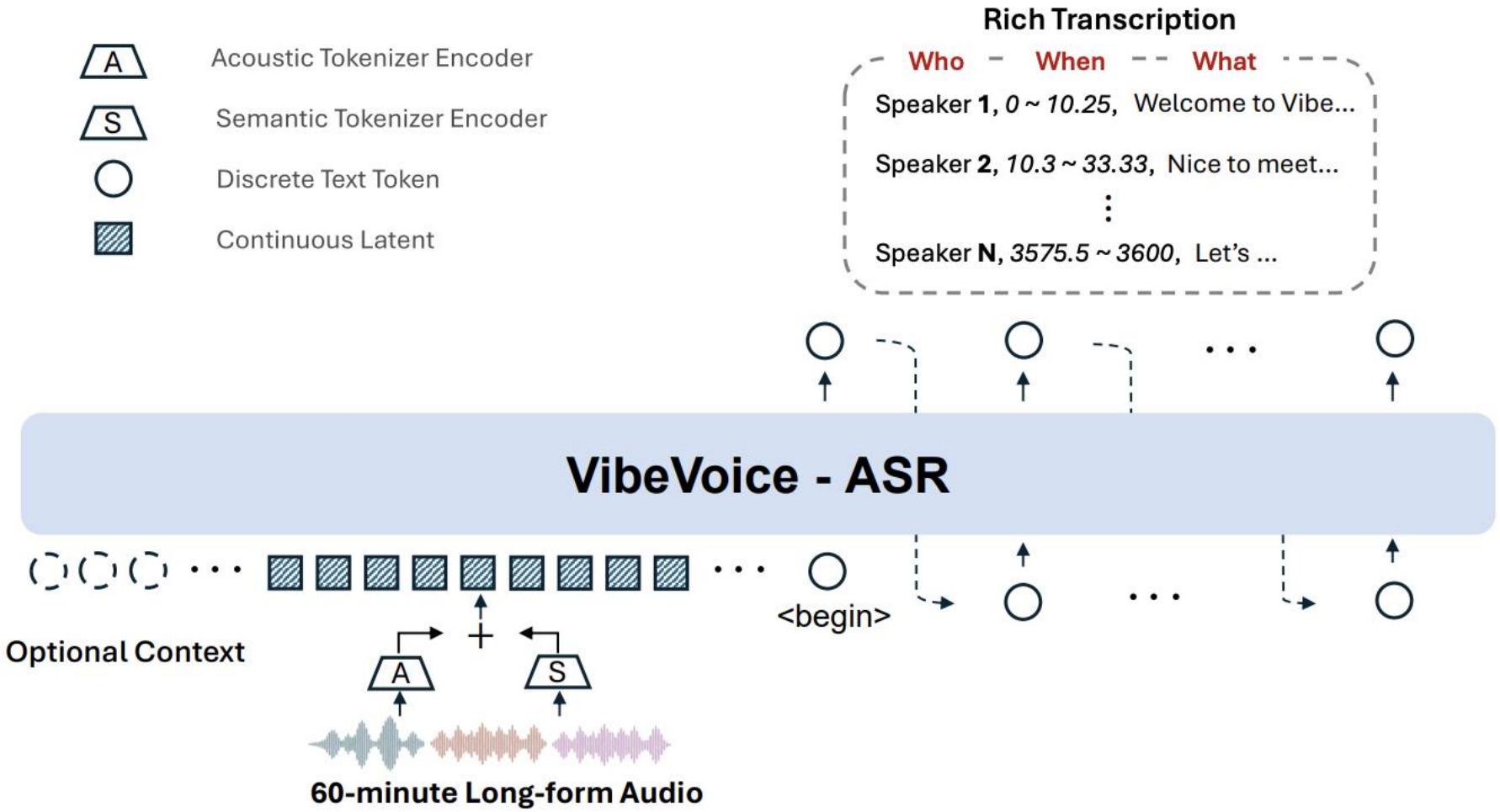


Figure 2: The architectural overview of VIBEVOICE-ASR. VIBEVOICE-ASR processes 60-minute long-form audio in a single pass by ingesting continuous latents from dual-tokenizers alongside optional user-provided context. The output is a generated stream of Rich Transcription, explicitly interleaving Speaker ID (Who), Timestamps (When), and Content (What)

MiMo-Audio: Audio Language Models are Few-Shot Learners-2025

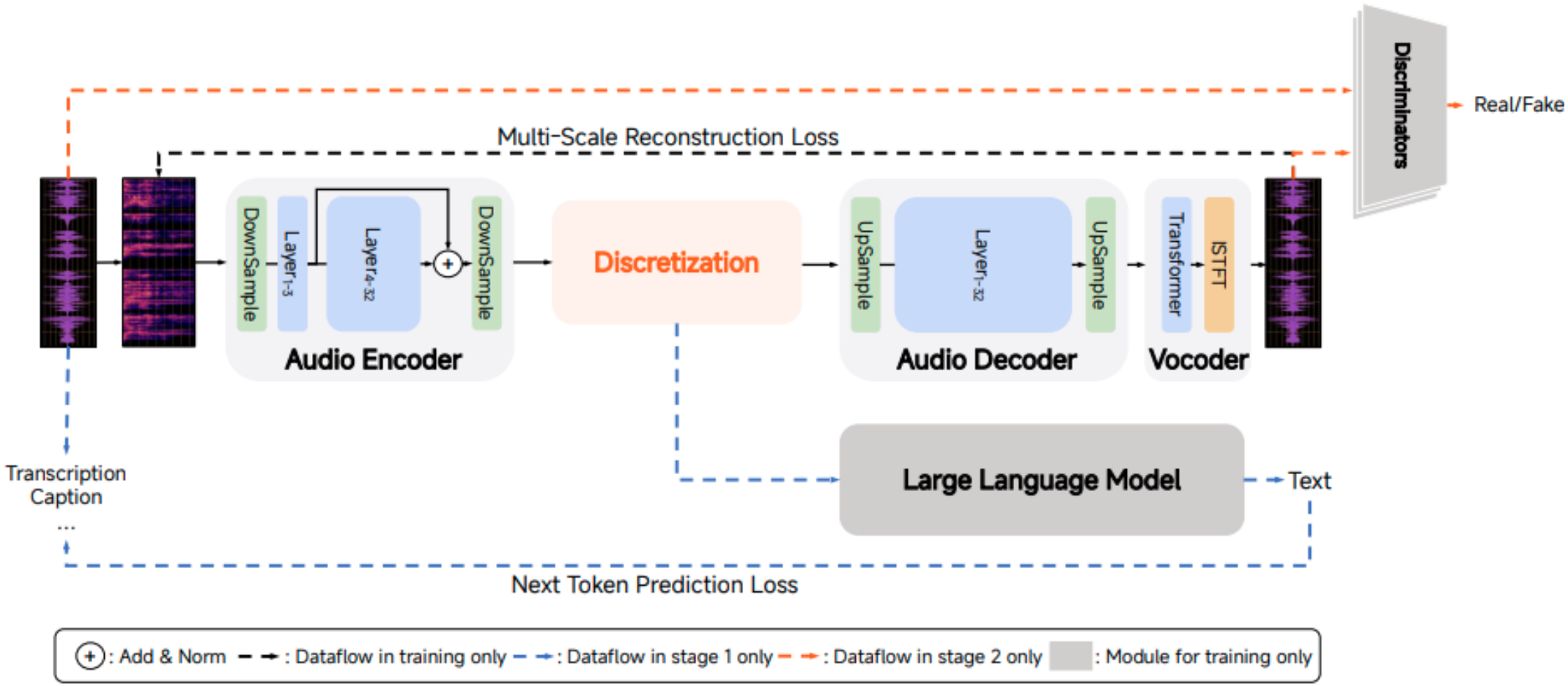


Figure 2 Illustration of MiMo-Audio-Tokenizer framework.

Step-Audio 2 Technical Report 27 Aug 2025

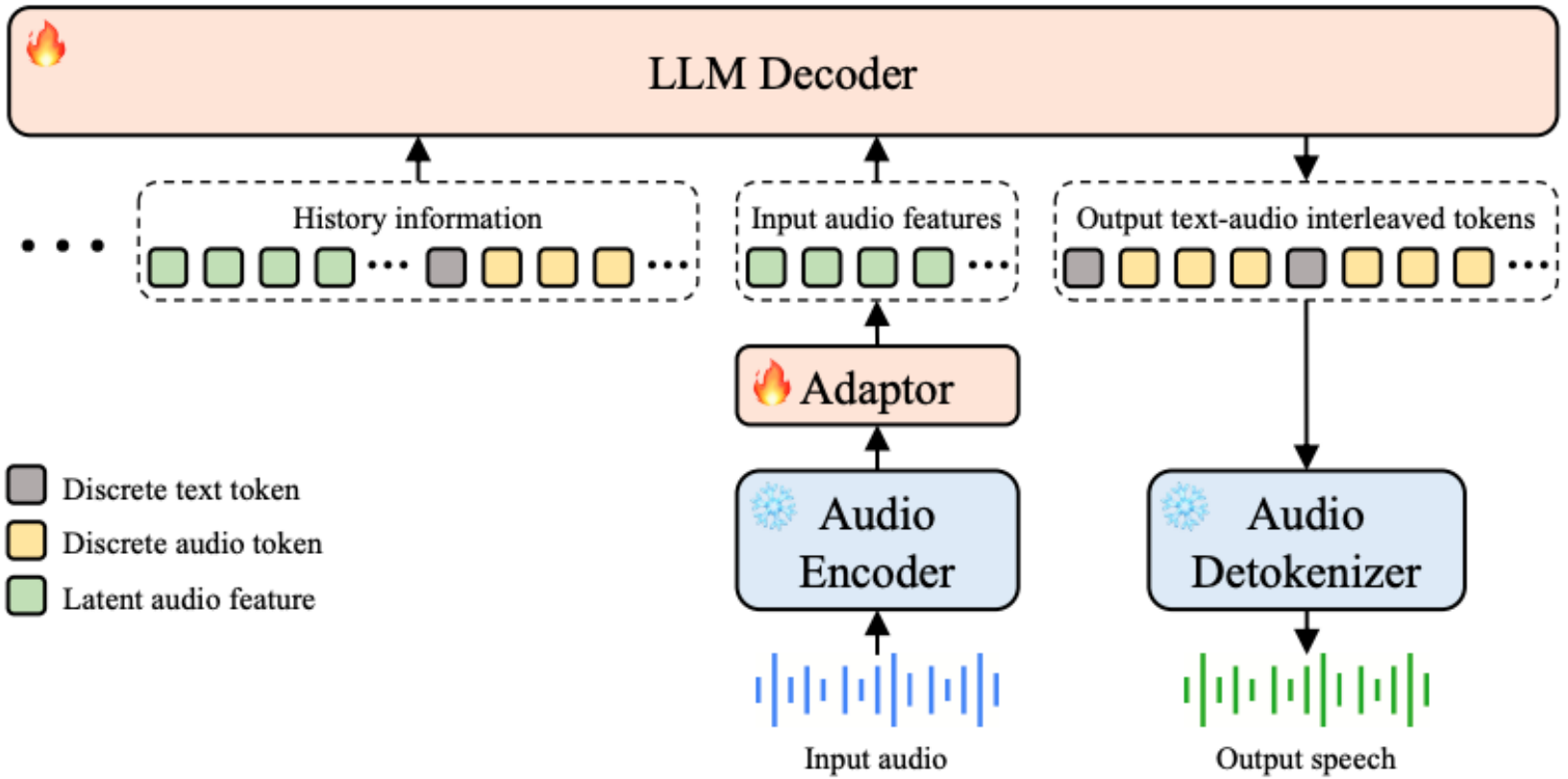


Figure 3: Architecture of the Step-Audio 2.

MOSS-Audio Technical Report 2026.07

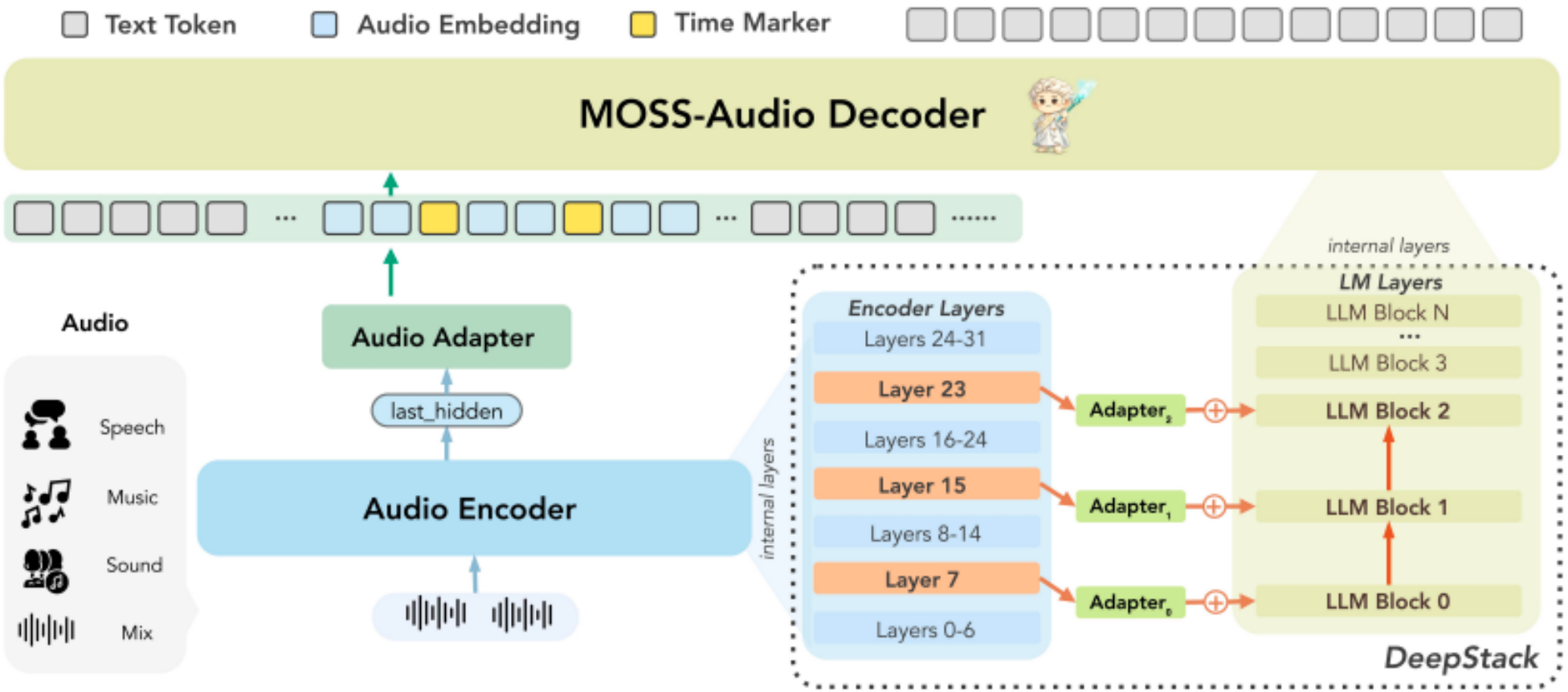


Figure 2 Architecture of MOSS-Audio.

Step-Audio 2 Technical Report 27 Aug 2025

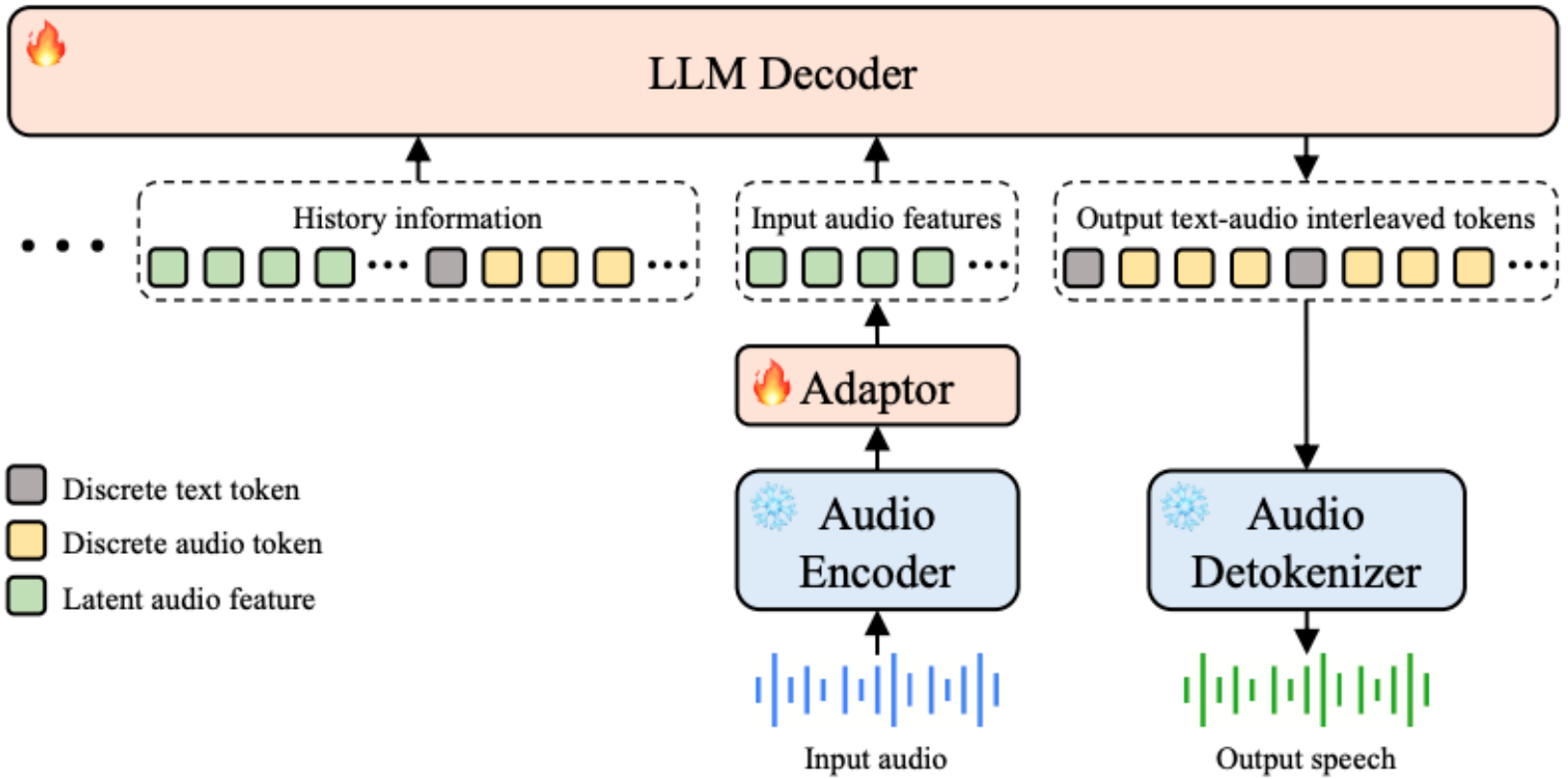


Figure 3: Architecture of the Step-Audio 2.

Qwen3-ASR Technical Report 2026-02-02

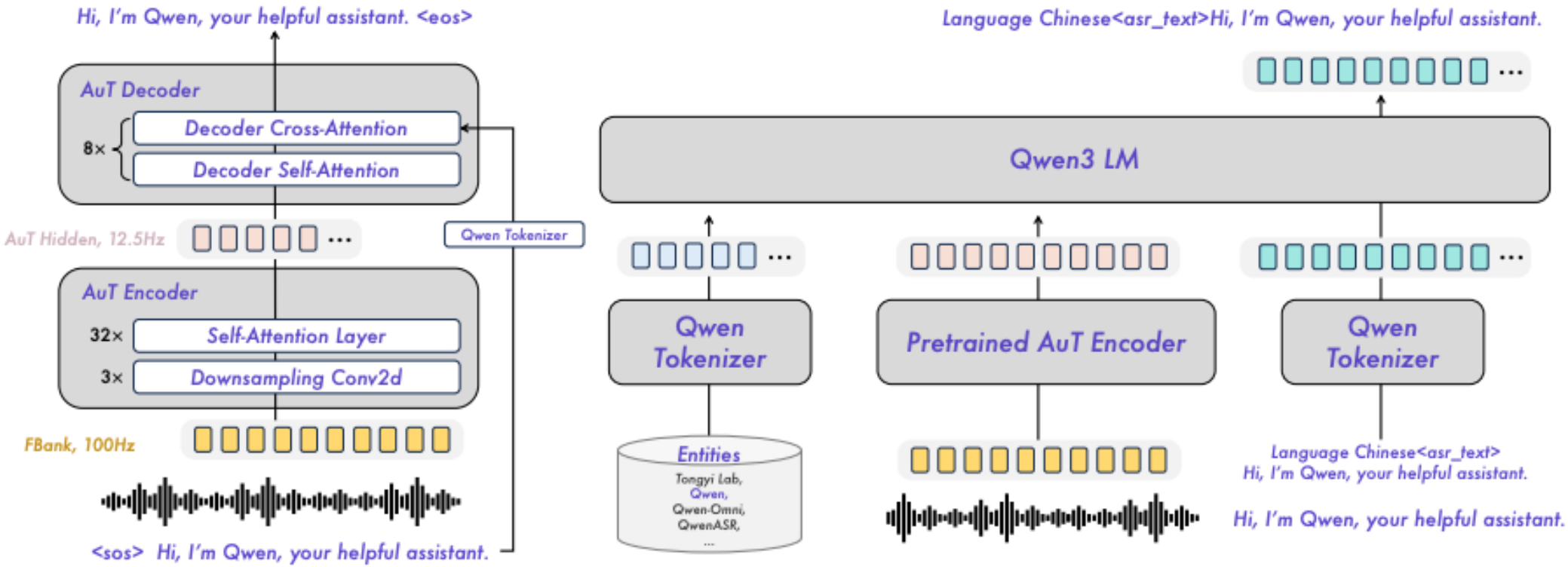


Figure 2: Architecture of AuT (left) and the overview of Qwen3-ASR (right).

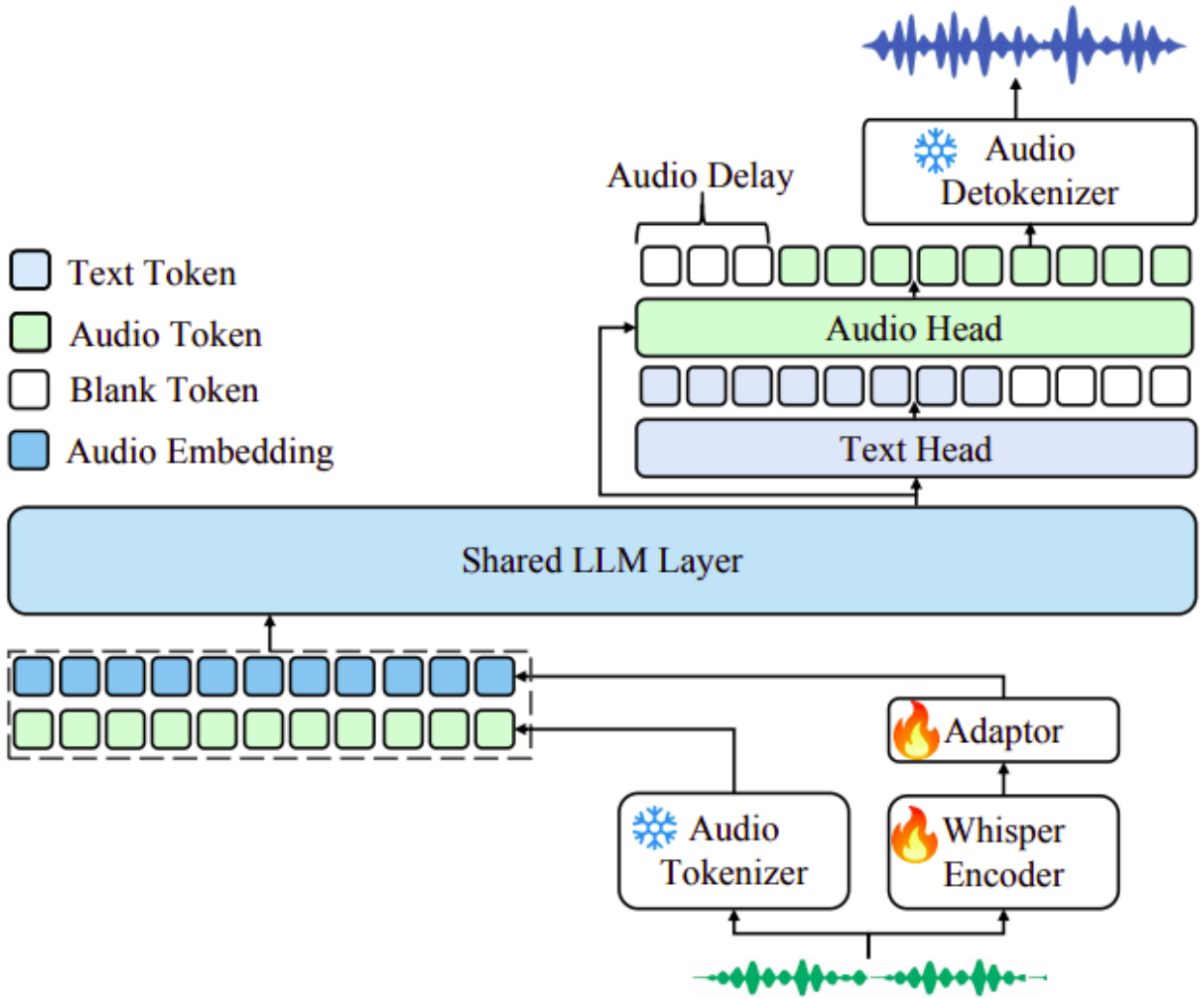


Figure 2: Overview of the Kimi-Audio model architecture: (1) an audio tokenizer that extracts discrete semantic tokens and a Whisper encoder that generates continuous acoustic features; (2) an audio LLM that processes audio inputs and generates text and/or audio outputs; (3) an audio detokenizer converts audio tokens into waveforms.